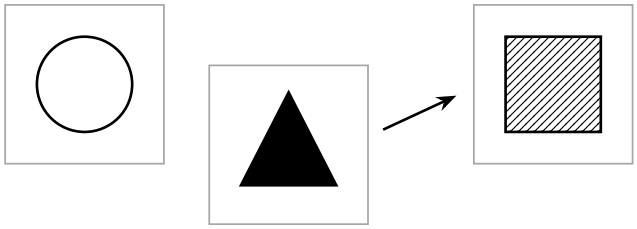
Use the nine tiles, one of each shape and fill. An allowed three uses three different tiles. Its shapes are all the same or all different. Its fills are also all the same or all different. Tile positions do not matter.



allowed three

shapes ○ △ □ all different

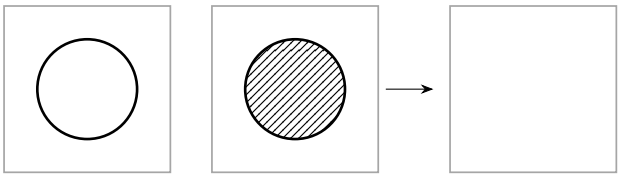
fills □ ■ ▨ all different

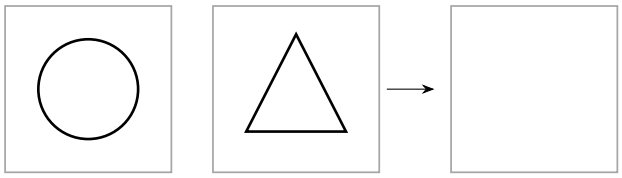
not allowed

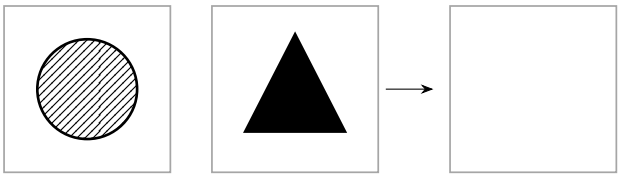
shapes ○ △ □ all different

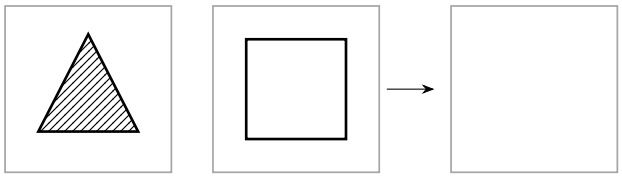
fills □ □ ■ two the same

Problem 1: Pick a third tile for each pair to make an allowed three. Then give your partner new pairs.

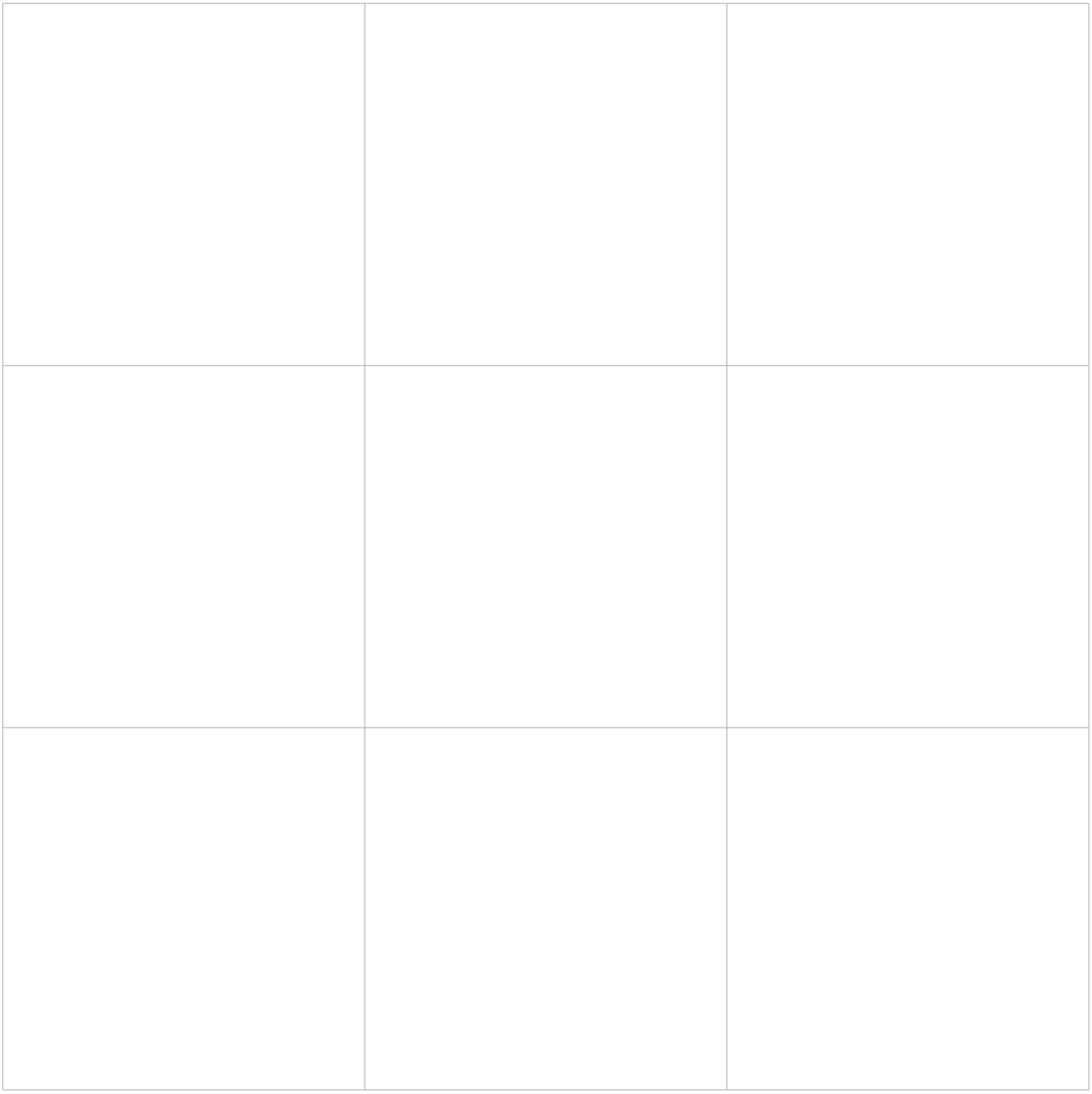




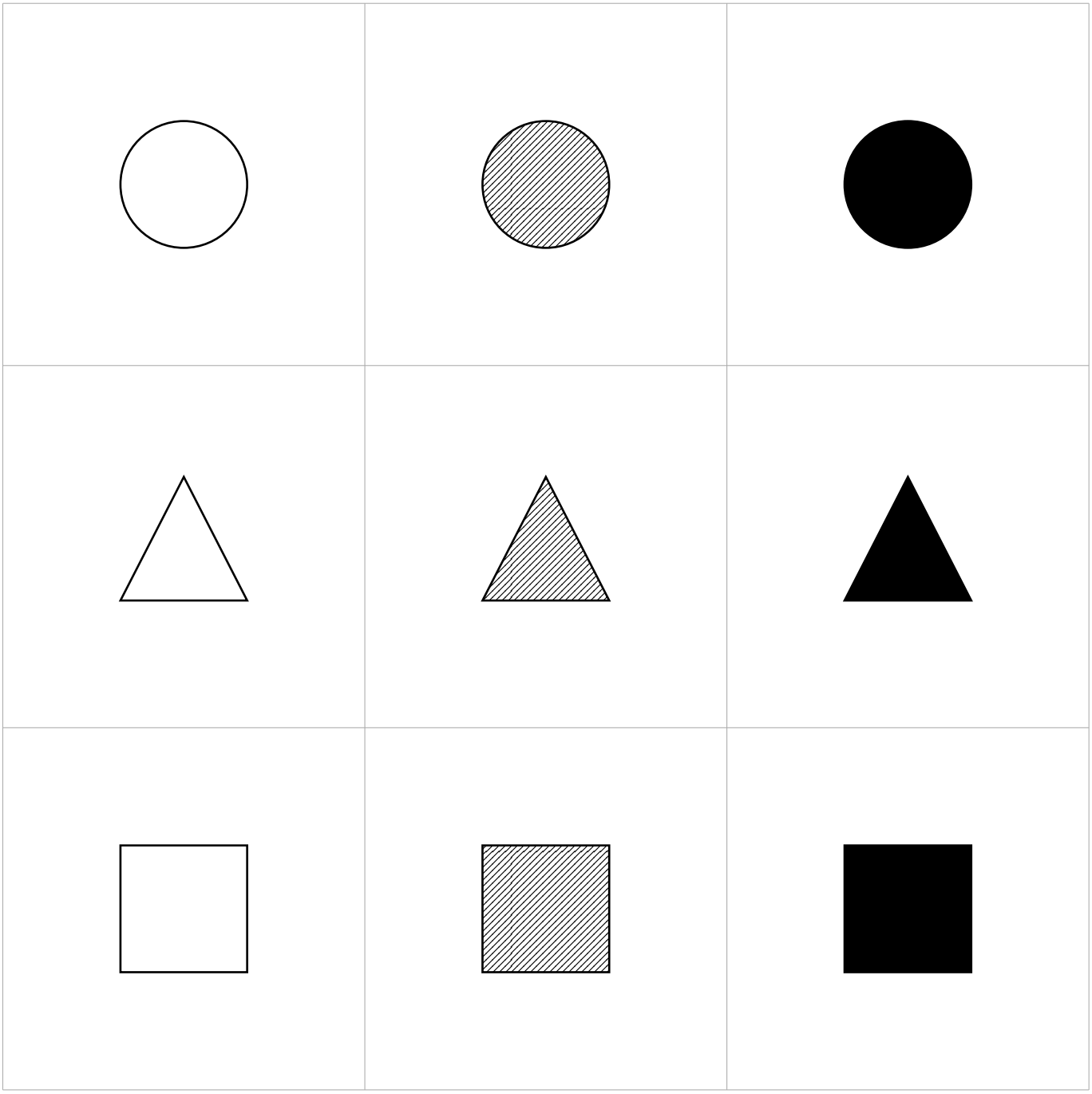




Problem 2: Put all nine tiles into three allowed threes. Find different ways.



Problem 3: Choose tiles without making any allowed three. How many can you keep?



Problem 4: Choose one tile and keep it in the middle of the table. Find all allowed threes using it; then try other middle tiles.